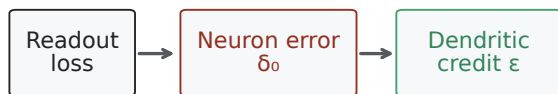


A

From a task error to a local update



$$\epsilon = A \Gamma c$$

A: address

 Γ : gain

c: coefficient

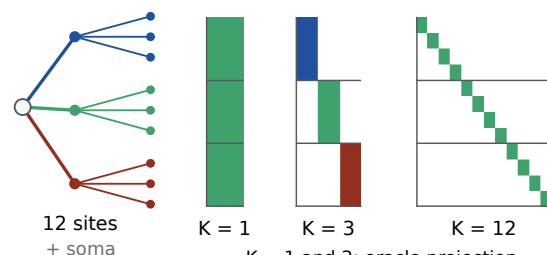
which sites

how strongly

which signal

Update = -learning rate $\times$ local eligibility $\times$ credit

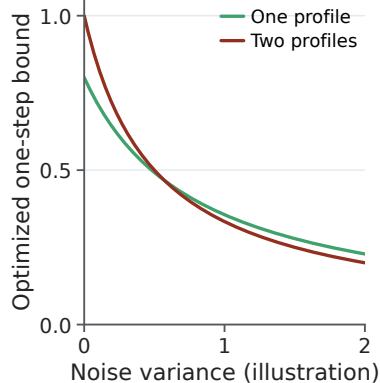
Error source and spatial delivery are separate choices

BMorphology supplies spatial profiles
Delivered field = $A c$ 

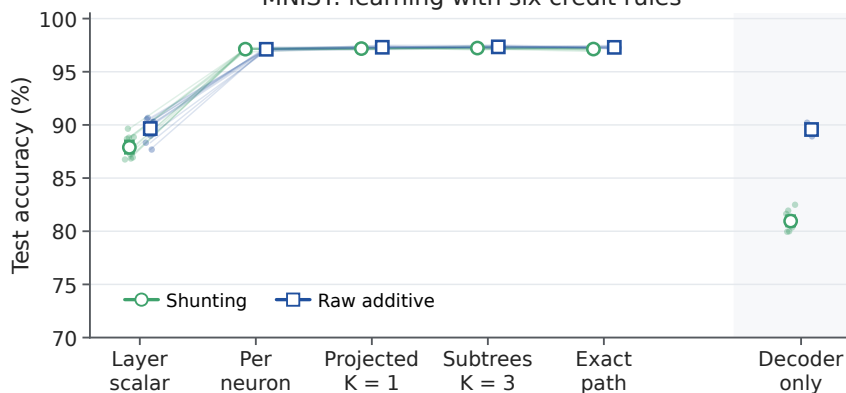
Rows: sites grouped by subtree; columns: profiles

C

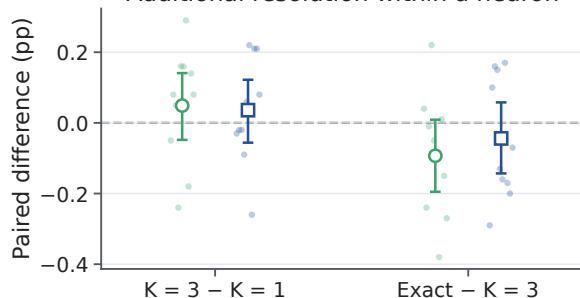
Noise and resolution

**D**

MNIST: learning with six credit rules

**E**

Additional resolution within a neuron

**F**

Separate image controls

DFA: shunt

DFA: additive

CIFAR: additive

